

Final project

Dr. Florian Hastreiter

Project start: Do, 20.08.2026

Display week: 1

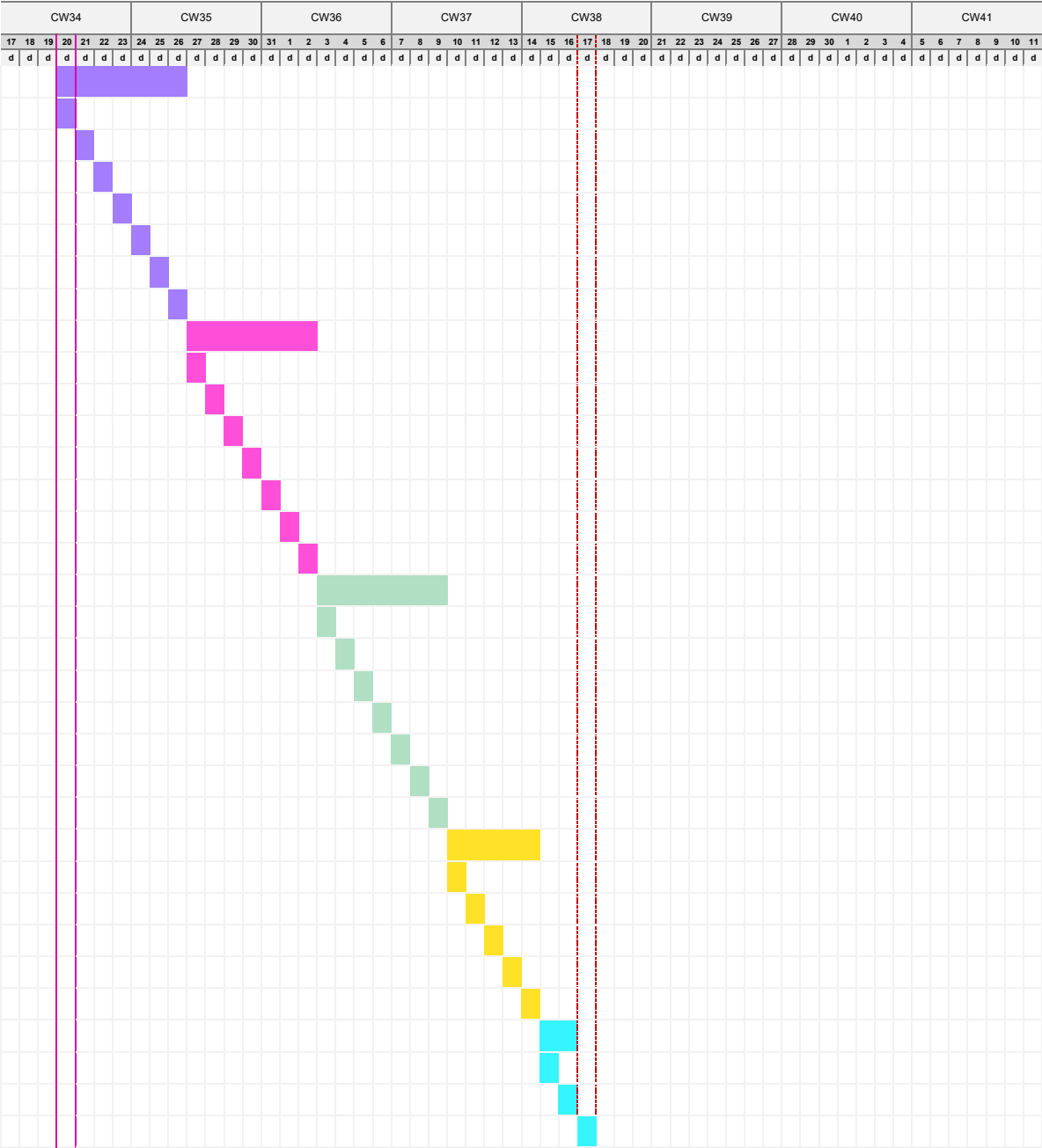
WBS Coding School

Explainable Machine Learning for Anomaly Detection and Fault Diagnosis in Batch Distillation

Due Date: 17.09.2026

Days Left: 28

| TASK                                  | DELIVERABLE                              | PROGRESS | DURATION | START      | END        |
|---------------------------------------|------------------------------------------|----------|----------|------------|------------|
| 1. Data & Problem Definition          | M1 – Data Ready                          | 0%       | 6        | 20.08.2026 | 26.08.2026 |
| Project setup & dataset download      | Data available, project structure        | 0%       | 1        | 20.08.2026 | 20.08.2026 |
| Dataset documentation & structure     | Understand files, experiments, variables | 0%       | 1        | 21.08.2026 | 21.08.2026 |
| Sensors & actuators analysis          | Variable overview                        | 0%       | 1        | 22.08.2026 | 22.08.2026 |
| Anomaly labels & metadata             | Understand target/fault structure        | 0%       | 1        | 23.08.2026 | 23.08.2026 |
| Data quality analysis                 | Missing values, duplicates, outliers     | 0%       | 1        | 24.08.2026 | 24.08.2026 |
| Define ML problem & evaluation strate | Target + train/test strategy             | 0%       | 1        | 25.08.2026 | 25.08.2026 |
| Define ML problem & evaluation strate | M1 – Data Ready                          | 0%       | 1        | 26.08.2026 | 26.08.2026 |
| 2. EDA & Feature Engineering          | M2 – First Model                         | 0%       | 6        | 27.08.2026 | 02.09.2026 |
| Sensor EDA                            | Distributions & patterns                 | 0%       | 1        | 27.08.2026 | 27.08.2026 |
| Normal vs. anomaly analysis           | Comparative analysis                     | 0%       | 1        | 28.08.2026 | 28.08.2026 |
| Time-series analysis                  | Temporal patterns                        | 0%       | 1        | 29.08.2026 | 29.08.2026 |
| Correlation & feature analysis        | Feature relationships                    | 0%       | 1        | 30.08.2026 | 30.08.2026 |
| Preprocessing pipeline                | Clean data pipeline                      | 0%       | 1        | 31.08.2026 | 31.08.2026 |
| Feature engineering                   | ML features                              | 0%       | 1        | 01.09.2026 | 01.09.2026 |
| Baseline model                        | M2 – First Model                         | 0%       | 1        | 02.09.2026 | 02.09.2026 |
| 3. ML Development                     | M3 – Model Freeze                        | 0%       | 6        | 03.09.2026 | 09.09.2026 |
| Baseline model comparison             | Initial benchmark                        | 0%       | 1        | 03.09.2026 | 03.09.2026 |
| Random Forest / tree models           | Advanced baseline                        | 0%       | 1        | 04.09.2026 | 04.09.2026 |
| Gradient Boosting / XGBoost           | Candidate model                          | 0%       | 1        | 05.09.2026 | 05.09.2026 |
| Cross-validation & imbalance          | Reliable evaluation                      | 0%       | 1        | 06.09.2026 | 06.09.2026 |
| Hyperparameter tuning                 | Optimized candidates                     | 0%       | 1        | 07.09.2026 | 07.09.2026 |
| Error analysis                        | Understand failures                      | 0%       | 1        | 08.09.2026 | 08.09.2026 |
| Final model selection                 | M3 – Model Freeze                        | 0%       | 1        | 09.09.2026 | 09.09.2026 |
| 4. Explainability & Validation        | M4 – Analysis Freeze                     | 0%       | 4        | 10.09.2026 | 14.09.2026 |
| SHAP implementation                   | Explainability pipeline                  | 0%       | 1        | 10.09.2026 | 10.09.2026 |
| Global feature importance             | Main drivers identified                  | 0%       | 1        | 11.09.2026 | 11.09.2026 |
| Local explanations                    | Individual predictions                   | 0%       | 1        | 12.09.2026 | 12.09.2026 |
| Fault/engineering interpretation      | Compare ML vs. known faults              | 0%       | 1        | 13.09.2026 | 13.09.2026 |
| Final validation & visualizations     | M4 – Analysis Freeze                     | 0%       | 1        | 14.09.2026 | 14.09.2026 |
| 5. Finalization                       | M5 – Project Freeze                      | 0%       | 1        | 15.09.2026 | 16.09.2026 |
| Code + documentation + README         | Project finalized                        | 0%       | 1        | 15.09.2026 | 15.09.2026 |
| Presentation + rehearsal              | M5 – Project Freeze                      | 0%       | 1        | 16.09.2026 | 16.09.2026 |
| Final presentation                    | FINAL                                    | 0%       | 1        | 17.09.2026 | 17.09.2026 |



| Days Left: 28                  |             |          |          |       |     |
|--------------------------------|-------------|----------|----------|-------|-----|
| TASK                           | DELIVERABLE | PROGRESS | DURATION | START | END |
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| CW34 |    |    |    |    | CW35 |    |    |    |    | CW36 |    |    |    |    | CW37 |   |   |   |   | CW38 |   |   |   |    | CW39 |    |    |    |    | CW40 |    |    |    |    | CW41 |    |    |    |    |    |    |    |    |    |   |   |   |   |   |   |   |   |   |    |    |   |
|------|----|----|----|----|------|----|----|----|----|------|----|----|----|----|------|---|---|---|---|------|---|---|---|----|------|----|----|----|----|------|----|----|----|----|------|----|----|----|----|----|----|----|----|----|---|---|---|---|---|---|---|---|---|----|----|---|
| 17   | 18 | 19 | 20 | 21 | 22   | 23 | 24 | 25 | 26 | 27   | 28 | 29 | 30 | 31 | 1    | 2 | 3 | 4 | 5 | 6    | 7 | 8 | 9 | 10 | 11   | 12 | 13 | 14 | 15 | 16   | 17 | 18 | 19 | 20 | 21   | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |   |
| d    | d  | d  | d  | d  | d    | d  | d  | d  | d  | d    | d  | d  | d  | d  | d    | d | d | d | d | d    | d | d | d | d  | d    | d  | d  | d  | d  | d    | d  | d  | d  | d  | d    | d  | d  | d  | d  | d  | d  | d  | d  | d  | d | d | d | d | d | d | d | d | d | d  | d  | d |